



An examination of the daily relationship between job boredom and later burnout and turnover intentions along with mitigating strategies

JeongJin Kim¹ · Seth A. Kaplan¹ · John A. Aitken¹ · Lida Ponce¹

Accepted: 2 April 2025 / Published online: 11 April 2025

This is a U.S. Government work and not under copyright protection in the US; foreign copyright protection may apply 2025

Abstract

Job boredom is one of the most prevalent aversive states experienced in the workplace. Although researchers have examined the impact of job boredom on important workplace outcomes, the day-to-day experience of job boredom and individual strategies to reduce its negative influence remain largely unexplored. Therefore, we examined the within-person relationships of job boredom with daily job burnout and turnover intentions, as well as two potential moderators to alleviate the impact of job boredom—psychological detachment from work during work hours and perceived cyberloafing norms. Using a daily diary study over 10 consecutive workdays (three surveys per day), we sampled 1,160 daily observations nested within 120 individuals. Hypotheses were tested using multilevel modeling while controlling for previous-day (lagged) outcome variables. Our results showed that job boredom was positively related to job burnout at the within-person level while controlling for previous-day job burnout. However, job boredom was not related to turnover intentions at the within-person level. Neither psychological detachment nor perceived cyberloafing norms attenuated the relationships of job boredom with job burnout or turnover intentions; thus, our moderation hypotheses were not supported. Rather, contrary to our prediction, we found that psychological detachment reinforced the relationship between job boredom and job burnout. We discuss the implications of our findings, study limitations, and future research directions.

Keywords Job boredom · Job burnout · Turnover intentions · Cyberloafing norms · Psychological detachment

Boredom is one of the most prevalent affective states in everyday life (Chin et al., 2017). It is pervasive among people of different ages, genders, and cultures (Weinstein et al., 1995), across the domains of work, school, and home (Chin et al., 2017; Mael & Jex, 2015), and across professions (Brodsky & Amabile, 2018; Melamed et al., 1995). At work, boredom is known to be detrimental to employees and organizations in many ways. Research estimates an annual financial loss of over \$100 + billion dollars attributed to 7.4 billion hours of idle time per year experienced by the U.S. workforce (“a period of involuntary downtime during which in-role tasks cannot be done,” Brodsky & Amabile, 2018, p. 496), during which people feel bored (Brodsky & Amabile, 2018; Zeschke & Zacher, 2023). Findings like this one are consistent with research showing that job boredom

relates to higher turnover intentions (Harju et al., 2014), job dissatisfaction (Melamed et al., 1995), and counterproductive work behaviors (CWB; Bruursema et al., 2011; Kim et al., 2021, 2024).

Despite the relevance and impact of boredom at work, boredom remains among the least understood form of affect in the workplace (Loukidou et al., 2009). More practically, there is little organizational research on how people experience discrete episodes of job boredom and how workers behave during, and respond to, such episodes. In addition, the limited relevant work on job boredom and alleviating the effects of job boredom has primarily been at the between-person level, which depicts boredom as a state that some feel more frequently than others (e.g., Game, 2007; Pindek et al., 2018). Between-person studies generally support the negative impact of job boredom (Fisher, 1993; Loukidou et al., 2009; van Hooff & van Hooff, 2014). However, although there have been some within-person studies on job boredom (e.g., Kim et al., 2024; van Hooff & van Hooff, 2016, 2017, 2023; Zeschke & Zacher, 2023), knowledge regarding job boredom and adaptive responses to those experiences is

✉ JeongJin Kim
jkim307@gmu.edu

¹ Department of Psychology, George Mason University,
Fairfax, VA, USA

still lacking at the within-person level (for an exception, see Spanouli et al., 2023). This is surprising given that boredom is an affective state whose intensity fluctuates vastly over days, weeks, and so forth (Baratta & Spence, 2018).

As indicated by between-person findings in job boredom research, we submit that knowledge of individuals' within-person experience of job boredom—that is, the manner in which boredom can fluctuate from one moment to another for a given person during the workday—can bring important practical implications. In particular, we contend that daily fluctuations of job boredom relate to daily fluctuations in two outcomes, including job burnout (i.e., a stress response characterized by cynicism, emotional exhaustion, and inefficacy; Maslach et al., 2001) and turnover intentions (i.e., individuals' intentions to quit their current jobs; Lee et al., 2014). Job burnout and turnover intentions have been highlighted as alarming workplace issues to both employees and organizations (Kelloway et al., 2023). Moreover, past studies show that there is substantial within-person variability in burnout and turnover intentions (approximately 21–47%), indicating that both constructs fluctuate over a short timeframe (Blanco-Donoso et al., 2021; Shi et al., 2021). Although both burnout and turnover intentions show within-person variability, the literature has not provided any theories or empirical evidence regarding the within-person effect of job boredom on these constructs.

Thus, based on theories of boredom and related empirical evidence, here, we study within-person effects of job boredom on burnout and turnover intentions, and individuals' strategies to mitigate the effects of job boredom. By doing so, we intend to provide several theoretical contributions and to offer practical implications. First, we contribute to the literature on boredom by examining within-person associations of job boredom with job burnout and turnover intentions. Specifically, we position these variables at the daily level to study the within-person, day-level relationships among them. Doing so is important because boredom represents an intrapsychic affective state (Eastwood et al., 2012; Fisher, 1993) that fluctuates within individuals over time (Baratta & Spence, 2018; Spanouli et al., 2023; van Hooft & van Hooft, 2017; Hooft & Hooft, 2023). As such, states of boredom derive from, interact with, and result in, other temporally adjacent phenomena (Westgate & Wilson, 2018). Notably, this characterization differs from the one underlying most studies of job boredom, which have treated it as a between-person variable, correlating differences in job boredom with job-related predictors and outcomes (e.g., Harju et al., 2014; Kim et al., 2021; van Hooft & van Hooft, 2014). Boredom does not necessarily function in the same manner at these two levels of analysis, though. For instance, being bored when one is typically not bored (i.e., a within-person deviation) plausibly functions quite differently than

does regularly experiencing high levels of boredom (e.g., between-person average boredom). In the former case, individuals may find such an experience especially unpleasant or, alternatively, a welcome change. In the latter case, research clearly shows that high levels of boredom are associated with various negative individual and organizational outcomes (Harju et al., 2014; Kim et al., 2021; Melamed et al., 1995). As this example illustrates, testing how boredom functions at the within-person level is important for testing the homology (i.e., equivalence) of findings and corresponding theory established at the between-person level (see McCormick et al., 2020). To the best of our knowledge, the current study is among the first to examine the association between job boredom and burnout and turnover intentions within a relatively short time frame (i.e., daily level).

A second theoretical contribution of our study involves providing knowledge about when and for whom the within-person effects of job boredom on the outcomes may vary (i.e., moderating conditions). Drawing from the meaning and attentional components (MAC) model of boredom (Westgate & Wilson, 2018) and its proposed routes to alleviating the harmful effects of boredom (i.e., regulating mental resources and switching activities), we study two potential moderators—psychological detachment during work hours and perceived cyberloafing norms at work—the former at the within-person level and the latter at the between-person level. Here, we attempt to extend the limited research on coping mechanisms or responses to boredom (e.g., boredom coping in Game, 2007; cognitive reappraisal in Spanouli et al., 2023). Our objective in studying these moderators is to offer practical implications and recommendations for human resources management practices regarding how to mitigate boredom's negative downstream effects on a daily basis.

In the remainder of the Introduction section, we review relevant literature and develop hypotheses based on the literature review. We then present an experience sampling method (ESM) study examining the within-person phenomena of job boredom, examining how daily episodes of job boredom relate to daily job burnout and daily turnover intentions, and potential moderators that might attenuate the negative impact of job boredom.

A review of boredom

Over the last two decades, there has been a surge in the study of boredom across various areas of psychology, such as industrial-organizational, social, educational, developmental, and clinical psychology, as well as in other disciplines, including animal science, economics, neuroscience, philosophy, sociology, and anthropology (Elpidorou, 2017).

Some of this recent work has sought to illuminate the experience of boredom and provide insights into its antecedents and consequences.

From this research, there is a consensus that boredom can be defined as a transient and unpleasant affective state that can both activate and deactivate individuals' cognition and behavior (Fisher, 1993; Watson & Tellegen, 1985), such as helping behavior (van Tilburg & Igou, 2017) and creativity (Park et al., 2019; Velasco, 2017). The feelings of boredom can be explained by three major accounts, including attentional/cognitive accounts (Eastwood et al., 2012; Smith & Ellsworth, 1985), functional/regulatory accounts (Bench & Lench, 2013, 2019; Elpidorou, 2017, 2018), and environmental accounts (Mikulas & Vodanovich, 1993).

More recently, Westgate and Wilson (2018) integrated the cognitive and functional accounts to propose the MAC model of boredom. Their model posits that boredom is a functional affective state that arises when people are unable (i.e., attentional deficits) or unwilling (i.e., meaning deficits) to cognitively engage with an activity that they value. That is, boredom has both attentional and meaningful components: boredom occurs when people (1) cannot focus on an activity and/or (2) do not find it meaningful. As such, boredom provides individuals a signal that they need to find a more interesting or meaningful activity than the current one. In addition, Westgate and Wilson (2018) discuss four strategies to assuage boredom: switching activities (i.e., doing something else rather than the current task), regulating goal value (i.e., modifying, switching, or reconstruing goals), regulating cognitive demands (i.e., regulating the demands of the current task), and regulating mental resources (i.e., increasing mental resources). These strategies are further discussed in terms of different profiles of boredom in Westgate and Wilson (2018), which is beyond the scope of the current study.

Job boredom, job burnout, and turnover intentions

Consistent with research showing that boredom is an aversive affective state (Fisher, 1993), several cross-sectional studies reveal that job boredom negatively impacts an array of workplace health and well-being outcomes. These negative consequences of boredom include poor mental health (Fisher, 1993), poor subjective health (Harju et al., 2014), low job satisfaction (Kass et al., 2001), heightened job stress (Wan et al., 2014), and increased stress symptoms (Harju et al., 2014). As one illustrative study of this research, Harju et al. (2014) studied correlates of job boredom using a sample of 11,000 + employees from 87 Finnish organizations. They found that employees who felt more bored at work were

more likely to show poor self-rated health and stress symptoms than those who felt less bored at work.

Longitudinal research on boredom (e.g., ESM designs) has also shown the negative impact of job boredom on well-being outcomes at the within-person level of analysis. For example, van Hooff and van Hooft (2016) used a 5-day diary study and found that job boredom in the afternoon was positively related to negative mood in the evening. Using a 5-day diary study, van Hooff and van Hooft (2017) showed that job boredom on a given day negatively influenced intrinsic motivation on the next day via negative pre-work attitudes on that day. Using a five-wave longitudinal study across 12 months, Zeschke and Zacher (2023) showed that job boredom was related to lower job satisfaction within a given individual.

In the current study, we examine the relationship between job boredom and two outcomes, job burnout and turnover intentions. Job burnout refers to “a state of mental and physical exhaustion caused by one's professional life” (Freudenberger, 1974), and is theorized to be three dimensions: emotional exhaustion (i.e., “a lack of energy and a feeling that one's emotional resources are used up”), depersonalization (i.e., “the treatment of clients as objects rather than people” which may manifest among employees as, for example, cynicism toward coworkers), and lack of personal accomplishment (i.e., “a tendency to evaluate oneself negatively”; Cordes & Dougherty, 1993, p. 623). Job burnout is known to lead to various detriments to employees and organizations, including absenteeism, turnover, impaired productivity, and mental illnesses (Bakker et al., 2023; Cordes & Dougherty, 1993).

We use the lens of the Effort-Recovery (E-R) model (Meijman & Mulder, 1998), which offers a valuable perspective for comprehending the relationship between job boredom and job burnout. The model posits that individuals at work use personal resources (i.e., both physical and psychological resources) to address, or put *effort* into, daily work-related demands. Experiencing demanding work situations depletes personal resources and leads to chronic health issues. When an individual is bored at work, the resource depletion associated with boredom can be mentally draining (Bakker et al., 2011). Although this draining effect of boredom may appear counterintuitive, a significant body of research shows that boredom can have psychosomatic consequences, such as fatigue (Cummings et al., 2016). This is consistent with the fact that boredom often positively relates to a range of other negative emotions, such as anger, depression, frustration, loneliness, and restlessness (Goetz et al., 2014). Given these associations of boredom with unpleasant emotional experiences, the resource-draining effect of job boredom appears to be an important aspect of the experience. Therefore, job

boredom's draining of personal resources should therefore lead one to feel burned out.

Indeed, prior studies at the between-person level have provided some empirical evidence for a positive relation between job boredom and burnout. Harju et al. (2014) found that employees with higher job boredom were more likely to experience stress symptoms, which are closely tied to one of the job burnout dimensions, emotional exhaustion (Gaines & Jermier, 1983). Several other studies also reported correlational evidence for the positive association between job boredom and burnout, providing correlation coefficients ranging from 0.42 to 0.55 (e.g., Reijseger et al., 2013; Vîrgă et al., 2022). Some time-lagged evidence also exists for this linkage. Sousa and Neves (2021) found that individuals who were bored at work were likely to experience emotional exhaustion and disengagement from work two weeks later, via ruminative thoughts. Harju et al. (2022) demonstrated that job boredom predicted job burnout six months later.

However, the literature provides little evidence of the boredom–burnout relationship over a short timeframe, or at the within-person level. As individuals' state of burnout can change over time, burnout researchers highlight the importance of the dynamic aspect of burnout, as compared to its static level (Dunford et al., 2012). Likewise, job boredom refers to a transient and episodic affective state that changes over a short period of time (Mael & Jex, 2015; Mikulas & Vodanovich, 1993). Given that job boredom is an aversive and unpleasant experience in and of itself associated with stress and ill-being (e.g., van Hooft & van Hooft, 2014; Wan et al., 2014), we predict a positive within-person relationship between job boredom and burnout.

Hypothesis 1. Job boredom is positively related to job burnout at the within-person level.

A second important outcome that may result from daily boredom is greater turnover intentions, which has important practical implications for organizations. Employee turnover theory (Lee et al., 2014) provides general support for the positive relationship between job boredom and turnover intentions. Research on voluntary turnover suggests that one of the predominant reasons employees quit the job pertains to low job satisfaction (Hom et al., 2017; Lee et al., 2014). That is, employees who feel dissatisfied with their jobs are likely to consider quitting.

Relatedly, the literature on job boredom links it very closely to job dissatisfaction both conceptually and empirically. O'Hanlon (1981) notes that job dissatisfaction, along with disinterest, constitutes the affective experience of monotony. Other boredom researchers incorporate the dissatisfaction component into their definition of boredom (e.g., Farmer & Sundberg, 1986; Harju et al., 2014; Mikulas

& Vodanovich, 1993; Reijseger et al., 2013). For example, Mikulas and Vodanovich (1993) define “boredom as a state of relatively low arousal and *dissatisfaction* [*italics added*], which is attributed to an inadequately stimulating situation” (p. 3). Indeed, empirical evidence suggests a strong negative correlation between job boredom and job satisfaction ($r = -.43$; e.g., Reijseger et al., 2013). Then, it follows that bored employees are likely to be dissatisfied and resource-depleted (Meijman & Mulder, 1998), leading to feelings of burnout at work.

We posit that employees' intentions to leave the current job/organization are partly a within-person phenomenon. One's decision to quit the job or leave the organization (or stay) is known to be a process that unfolds over many psychological mechanisms (Lee et al., 2014). As job embeddedness theory suggests that there are a myriad of reasons for turnover (Lee et al., 2014), one's intentions to vacate the job should vary over a short timeframe. Indeed, past daily diary studies reported intraclass correlation coefficients (ICCs) for turnover intentions to be in the range of 0.53 to 0.79 (e.g., Blanco-Donoso et al., 2021; Shi et al., 2021), indicating that much of the variance in turnover intentions varies at the daily level.

Past research has provided very few empirical tests of the within-person association between job boredom and turnover intentions. Such is perhaps not surprising given the paucity of findings even at the between-person level. Here, though, we propose that, as individuals' intention to quit fluctuates within-person (Blanco-Donoso et al., 2021; Shi et al., 2021), the daily fluctuations in turnover intentions should be influenced by individuals' day-to-day fluctuations of a negative affective state (i.e., job boredom). Given that employees experience negative well-being outcomes from the feeling of boredom at work (e.g., job dissatisfaction, stress symptoms; van Hooft & van Hooft, 2014; Wan et al., 2014), these negative outcomes are likely to increase their intentions to turnover. Consistent with our prediction, van Hooft and van Hooft (2017) found that job boredom on a given day was positively related to pre-work negative attitudes on the next day. Shi et al. (2021) found that employees' turnover intentions increased on days when their subjective well-being (operationalized as positive affect, negative affect, and job satisfaction) decreased, compared to other days when their subjective well-being increased. Therefore, we hypothesize that the relationship between job boredom and turnover intentions is positive at the within-person level.

Hypothesis 2. Job boredom is positively related to turnover intentions at the within-person level.

Psychological detachment and perceived cyberloafing norms for alleviating the effects of job boredom

Given these hypothesized effects, an important question involves how the negative effects of boredom may be mitigated. Consistent with the E-R model, the drained resources are restored when individuals encounter or are exposed to conditions that facilitate recovery. Relying on Westgate and Wilson's (2018) routes to alleviating the effects of boredom (i.e., regulating mental resources and switching activities), we suggest two potential strategies. First, we posit that *psychological detachment from work* (or psychological detachment)—defined as employees' recovery experience of mentally disconnecting themselves or "switching off" from work (Sonnentag et al., 2010)—should be beneficial in alleviating the effects of boredom on outcomes by regulating mental resources at work. Although the work recovery literature generally has focused on detachment during *non-work hours* (Sonntag & Fritz, 2007), a more recent stream of research emphasizes the importance of recovery activities during *work hours* via means such as having micro-breaks at work (e.g., Kim et al., 2015, 2017), or "short respite activities that are undertaken voluntarily on a need basis between series of task episodes" (Kim et al., 2017, p. 30). Here, we posit that bored employees who can detach from work during work hours feel less burned out and have reduced turnover intentions.

This notion is consistent both with the MAC model (Westgate & Wilson, 2018) and with evidence on the impact of micro-breaks from the workplace recovery literature. Given that, as the MAC model posits, individuals feel bored due to a lack of attention to and/or meaning in the current work task, detaching from the work task for a brief time period (i.e., a short mental break) would facilitate resource regulation. In this vein, the work recovery literature describes that a mental break from one's work task during work hours can provide an opportunity to restore energy. Consistent with this notion, research on micro-breaks at work suggests that individuals having small cognitive breaks at work are more likely to feel positive affect (Kim et al., 2015) and rejuvenate their day-to-day attentional resources despite the demanding nature of their jobs (Kim et al., 2017). This theoretical and empirical support suggests that bored employees who can take brief mental breaks from work on the job (i.e., psychological detachment) are likely to experience reduced burnout and decreased turnover intentions, compared to those whose job/work environments do not allow such mental breaks. Therefore, we predict that psychological detachment would lessen bored employees' later experiences of job burnout and heightened intentions to quit.

Hypothesis 3. The positive within-person relationship between job boredom and job burnout is moderated by psychological detachment, such that psychological detachment weakens the relationship.

Hypothesis 4. The positive within-person relationship between job boredom and turnover intentions is moderated by psychological detachment, such that psychological detachment weakens the relationship.

The second strategy to fight off job boredom pertains to the activity-switching route proposed by Westgate and Wilson (2018). Specifically, here, we study this in the context of cyberloafing, which refers to the voluntary use of the internet during office hours to surf nonwork-related websites for personal purposes (Lim, 2002; for a review, see Lim & Teo, 2024). In the face of the deleterious impact of boredom, switching the current task to a nonwork task, such as browsing the web, can be a feasible option to replenish resources for most individuals in the workplace (Lim, 2002). Although most prior research on cyberloafing has focused on the harmful effects of cyberloafing for employees and organizations (Lim, 2002), some studies have highlighted its positive impact. Of most relevance here, one cross-sectional study demonstrated that employees who were bored due to work underload were more likely to engage in cyberloafing (Pindek et al., 2018). Other studies also provide indirect evidence for the restorative role of cyberloafing. For example, cyberloafing was positively associated with positive affect (Lim & Chen, 2012) and recovery experiences including psychological detachment (Wu et al., 2020a), and relaxation (Wu et al., 2020b). These findings suggest that using the internet for nonwork purposes can help alleviate feelings of boredom at work.

In this study, we focus on the normative aspect of cyberloafing at work (i.e., perceived cyberloafing norms at work), rather than one's actual cyberloafing behavior. Research shows that perceived norms (or norm-based interventions) are influential in changing individuals' attitudes and behaviors (Miller & Prentice, 2016; Rimal & Real, 2003, 2005). Given this, the beneficial impact of cyberloafing is more likely when employees perceive that brief internet browsing is condoned in the workplace (Koay & Soh, 2018). This is in line with Stofferregen's (2003) concept of affordances, which refer to "relations between an animal [or person in this case] and its environment that have consequences for behavior" and "properties of the environment of an animal that have consequences for the animal's behavior" in the context of the ecological approach to perception and behavior (p. 117). Regarding perceived cyberloafing norms, strong anti-cyberloafing norms would diminish situational affordances for bored individuals to browse the internet, whereas lenient cyberloafing norms would create situational affordances for

them to do so. Therefore, we predict that perceived cyberloafing norms at work would buffer the harmful influence of boredom.

Hypothesis 5. The positive within-person relationship between job boredom and job burnout is moderated by perceived cyberloafing norms, such that perceived cyberloafing norms at the between-person level weaken the within-person relationship.

Hypothesis 6. The positive within-person relationship between job boredom and turnover intentions is moderated by perceived cyberloafing norms, such that perceived cyberloafing norms at the between-person level weaken the within-person relationship.

Method

Participants and procedure

The dataset used in this study is part of a larger data collection; thus, participants completed additional measures, aside from those detailed below, beyond the scope of the current study. We recruited full-time employees by posting an advertisement on Craigslist, an online community website (see <https://www.craigslist.org/about>) in multiple large cities in the United States. The advertisement contained information about the study procedures, eligibility criteria, and compensation, and was posted for approximately two weeks in June 2022. Interested participants were requested to email the researchers and were subsequently invited to complete the initial baseline survey, which included informed consent and measures of perceived cyberloafing norms and demographic variables.

In the initial survey, participants were first screened based on three eligibility criteria: (a) age (i.e., 18 + years old), (b) full-time employment status (i.e., 30 + work hours per week), and (c) work schedules (i.e., whether they had a traditional work schedule from 9:00 AM to 5:00 PM), the last of which was implemented to facilitate scheduling of daily surveys. To increase participants' response rates, we employed a commitment device in the initial survey that asked participants to explicitly commit to completing a minimum of 75% of the daily surveys (Gabriel et al., 2019). If respondents failed to commit to completing 75% or more of daily surveys, we kindly asked them to consider increasing their commitment to maximize the utility of their effort in participating in the research study. As a result, all participants chose to commit 75% or more of the daily surveys.

After completing the baseline survey, participants received three brief surveys per day for 10 consecutive workdays (i.e., two consecutive workweeks). Shortly after

starting each workday (at 10:00 a.m.), participants received the first daily survey (T1), which included a momentary measure of job boredom. The second daily survey (T2), which was sent approximately halfway through each workday (at 1:00 p.m.), included a momentary measure of psychological detachment from work in the time interval after participants reported their level of boredom. The third and final survey (T3), which was sent shortly before the end of each workday (at 4:00 p.m.), included day-level measures of job burnout and turnover intentions. To prevent bias, the order in which items were presented in each survey was randomized within each measure (Turk et al., 2018). Forced answering was not used given its potential negative effects on data quality (Sisichka et al., 2022). In exchange for participation, participants received \$5 for completing the initial baseline survey, \$1 for completing each daily survey (up to \$3 per day), and a weekly bonus of \$10 if they completed at least 75% (11 out of 15 surveys) of daily surveys for each of the two weeks of participation, totaling a maximum of \$55.

To ensure high data quality, we followed best practice recommendations for convenience sampling. Specific strategies we implemented included: posting the advertisement in major metropolitan areas (Antoun et al., 2016), including attention check items in the initial survey (e.g., "Please select *Never* for this item."; Meade & Craig, 2012), and tracking unique identifiers to assure only one survey was completed per participant (Mason & Suri, 2012). Additionally, to ensure that our study had sufficient statistical power for multilevel analysis, we followed Gabriel et al.'s (2019) sample size recommendations, which were based on their calculation of the mean sample size of the past ESM studies. Accordingly, we aimed to obtain a minimum Level 2 sample size of 83 and a minimum Level 1 sample size of 835 by administering three daily surveys for 10 consecutive workdays. This strategy of sampling for two weeks allowed us to capture a generally representative sample of a person's daily experiences (Wheeler & Reis, 1991).

Initially, 165 individuals satisfied our eligibility criteria in the initial survey. Of these, we excluded 12 participants who failed one or more of the three attention check items, and 33 participants who did not complete a minimum of two full days of daily surveys (i.e., six daily surveys). The final sample size was 1,160 observations nested within 120 individuals (response rate = 96.67%). Participants were 32.93 years old on average ($SD = 3.27$), 32.03% of whom were female. Most participants reported to have held their current jobs for 2–5 years (55.83%) or 5–9 years (41.67%). A majority of participants had obtained a bachelor's degree (55.56%) and were either Black/African American (58.17%) or White (40.52%).

Measures

We used a 5-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*) for all measures unless otherwise noted.

Within-Person variables (Level 1)

Job boredom Momentary job boredom was assessed using a 4-item measure (Park et al., 2019). On a 5-point Likert scale (1 = *not at all* to 5 = *extremely*), participants were asked to rate the extent to which they were feeling each of the following four emotion adjectives at the current moment: “bored,” “sluggish,” “dull,” and “lethargic.” We aggregated momentary job boredom assessed in T1 and T2 up to a morning measure of job boredom.¹ Within- and between-person omega coefficients were 0.74 and 0.99, respectively.

Psychological detachment Psychological detachment from work during work hours was measured using a 4-item measure (Sonnentag & Fritz, 2007). We asked participants to rate each item while thinking about the past 2–3 h at work. Example items include “I got a break from the demands of work” and “I distanced myself from work.” Within- and between-person omega coefficients were 0.64 and 0.98, respectively. Past research suggests that within-person reliability estimates ranging from 0.50 to 0.70 are acceptable in within-person research that uses scales with fewer items (e.g., Fisher et al., 2016).

Job burnout Job burnout was measured using a single-item measure (Shockley et al., 2021) by asking participants to indicate the degree to which they felt burned out at work today.

Turnover intentions Turnover intention was assessed using a 4-item measure (Shi et al., 2021). Example items include “Today, I thought of quitting my job” and “Today, I considered leaving this job for a new job.” Within- and between-person omega coefficients were 0.54 and 0.93, respectively.

Control variables

To test the main effect of job boredom on the outcome variables (i.e., Hypotheses 1 and 2), we included the respective previous-day (i.e., lagged) outcome. For instance, when testing the within-person relationship between job boredom and job burnout (Hypothesis 1), we included lagged job burnout as a control variable. Controlling for the effect of previous-day outcome allowed us to address common

method bias regarding the concurrent relationship between job boredom and the outcome variables and hence allows us to obtain a more accurate estimate of within-person effects (Gabriel et al., 2019).

Between-Person variables (Level 2)

Perceived cyberloafing norms We assessed perceived cyberloafing norms at work using a single-item measure (“It seems to be okay to just surf the internet while at work”). This item was taken from a scale created by Anandarajan et al. (2000) and used by Liberman et al. (2011).

Data analysis

We used multilevel modeling (MLM) to test our hypotheses, using the lme4 package (Bates et al., 2015) in R. To obtain accurate estimates of within-person slopes (González-Romá & Hernández, 2017), we used person-mean centering for Level 1 (within-person) predictors (i.e., job boredom and psychological detachment) and Level 1 covariates (i.e., lagged outcome variables). For the Level 2 variable (i.e., cyberloafing norm), we used grand-mean centering. We also included the person mean of job boredom as a Level 2 predictor to account for the possible between-person effect of job boredom. This allowed us to test whether the job boredom–outcome relationship varied across between- and within-person levels of analysis (González-Romá & Hernández, 2017). To test the main effect hypotheses (Hypotheses 1 and 2), we ran random coefficient models in which job burnout and turnover intentions were separately regressed onto job boredom, the Level 2 person mean of job boredom, and the respective previous-day outcome (job burnout or turnover intentions, respectively). We also specified the slope of job boredom to be random in those models. To test Hypotheses 3 and 4, psychological detachment and a product term (job boredom $\times$ psychological detachment) were added to the random coefficient models for Hypotheses 1 and 2 to test the significance of the interaction effect. Similarly, to test Hypotheses 5 and 6, perceived cyberloafing norms and a product term (job boredom $\times$ perceived cyberloafing norms) were added to the same models for Hypotheses 1 and 2. Missing data were handled via listwise deletion.

Results

Table 1 presents intra-class correlations, descriptive statistics, reliability estimates, and correlations among study variables. The proportions of within-person variance for the key variables were greater than 0.05 (Julian, 2001), warranting multilevel investigation.

¹ We also attempted to use just the T1 job boredom, but the conclusions were not changed. We decided to use the aggregate measure of job boredom to increase reliability.

Table 1 Descriptive statistics and correlations among study variables

Variable	M	SD _{between}	SD _{within}	2	3	4	5	6	7	8
1. Age	32.93	3.27		−0.05	0.26**	−0.12	−0.05	−0.04	0.02	−0.01
2. Gender	0.66	0.48			0.10	−0.10	0.23*	−0.01	0.01	0.01
3. Job tenure	3.40	0.60				−0.16	0.04	0.21*	0.16	0.11
4. Perceived cyberloafing norms	3.44	1.19					−0.01	−0.12	0.07	0.14
5. Job boredom	1.56	0.67	0.30				(0.72)	0.50***	0.61***	0.72***
6. Job burnout	2.41	1.31	0.86				0.11**	(0.50)	0.60***	0.53***
7. Turnover intentions	2.06	0.86	0.51				0.07*	0.14***	(0.58)	0.80***
8. Psychological detachment	1.84	0.91	0.45				0.12***	0.05	0.14***	(0.67)

Coefficients below the diagonal indicate within-person correlations while coefficients above the diagonal represent between-person correlations. Numbers along the diagonal in parentheses are intraclass correlations. 1. Gender was coded such that 0 = *female* and 1 = *male*. Job tenure was measured using an ordinal variable (1 = 1 year or less, 2 = 1 to 2 years, 3 = 2 to 5 years, 4 = 5 to 9 years, 5 = 10 years or more)

* $p < .05$, ** $p < .01$, *** $p < .001$

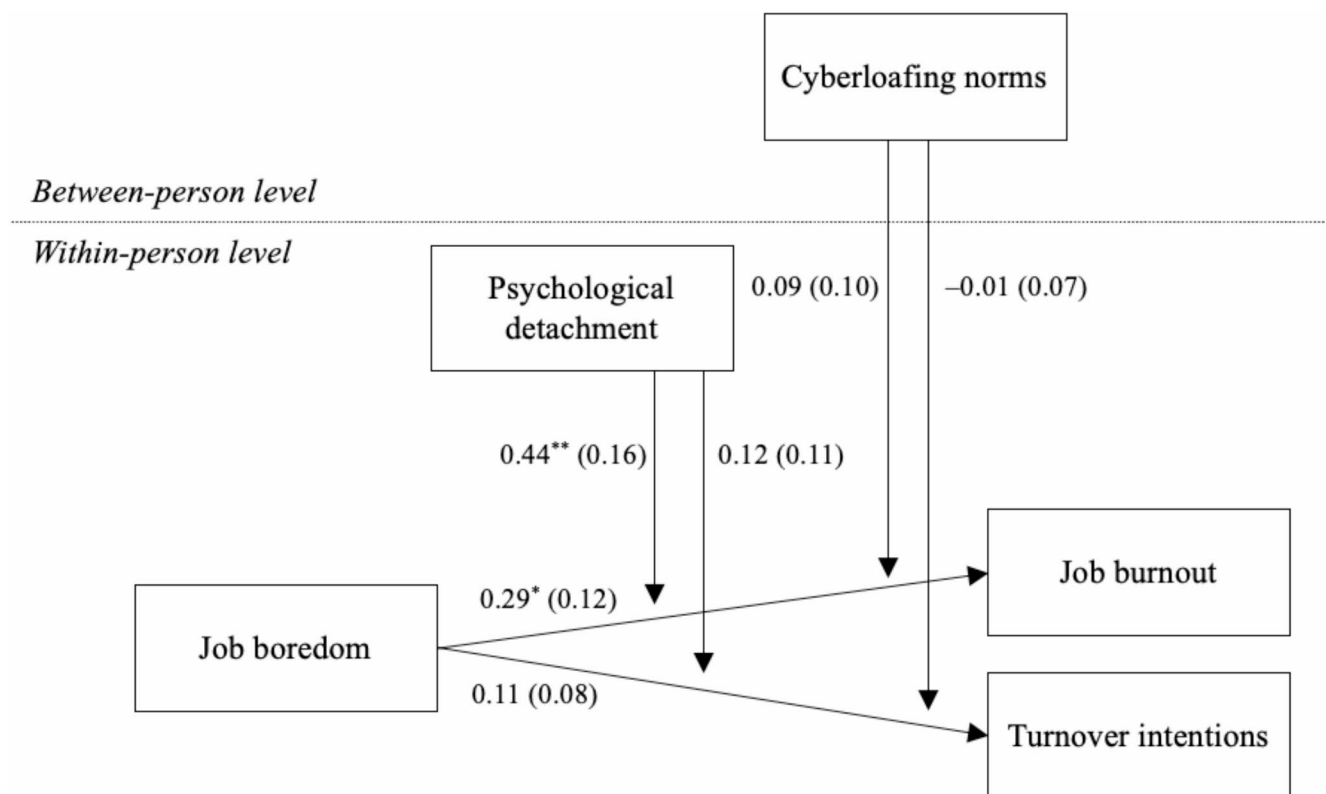


Fig. 1 The test model of the present study. *Note.* Unstandardized regression coefficients and standard errors (in parentheses) are reported. For simplicity, control variables are excluded graphically. * $p < .05$, ** $p < .01$

To ensure that the study variables were distinct, we performed multilevel confirmatory factor analyses (CFA) on three within-person variables. Job burnout and perceived cyberloafing norms were not included in the CFA model because they were measured using single-item measures. The results indicated good model fit ($\chi^2[102] = 202.524$, CFI = 0.972, TLI = 0.964, SRMR_{within} = 0.037, SRMR_{between} = 0.043, RMSEA = 0.034). These findings support the distinctiveness of our variables.

Figure 1 illustrates the study model with path coefficients, and Table 2 presents findings from the hypothesis tests. In

Hypothesis 1, we hypothesized that job boredom would be positively related to job burnout. As reported under Model 1 in Table 2, MLM results showed that job boredom was significantly and positively related to job burnout, while taking into account the respective lagged outcome ($\gamma = 0.29$, $p = .016$), supporting Hypothesis 1. In Hypothesis 2, we predicted that job boredom would be positively related to turnover intentions. As reported under Model 4 in Table 2, our findings showed that job boredom was not positively

Table 2 MLM results regarding the relationships of job boredom with job burnout and turnover intentions and moderation effects of psychological detachment and perceived cyberloafing norms

Variable	Job burnout (outcome)			Turnover intentions (outcome)		
	Model 1		Model 2		Model 3	
	γ (SE)	p	γ (SE)	p	γ (SE)	p
Constant	0.96 (0.24)	< 0.001	0.94 (0.24)	< 0.001	0.96 (0.24)	< 0.001
Lagged outcome (L1)	–0.07 (0.04)	0.092	–0.07 (0.04)	0.090	–0.07 (0.04)	0.091
Job boredom (L2)	0.91 (0.15)	< 0.001	0.92 (0.15)	< 0.001	0.91 (0.15)	< 0.001
Job boredom (L1)	0.29 (0.12)	0.016	0.22 (0.11)	0.056	0.30 (0.12)	0.015
Psychological detachment (L1)			0.01 (0.07)	0.902		
Job boredom $\times$ psychological detachment			0.44 (0.16)	0.009		
Perceived cyberloafing norms (L2)					–0.02 (0.07)	0.763
Job boredom $\times$ perceived cyberloafing norms					0.09 (0.10)	0.413
$Pseudo-R^2$	0.163		0.168		0.164	
					0.185	
					0.189	
					0.187	

$N = 739$ –740 observations nested within 107 individuals. L1 = Level 1 (within-person), L2 = Level 2 (between-person)

related to turnover intentions ($\gamma = 0.11$, $p = .146$).² Therefore, Hypothesis 2 was not supported.

In Hypothesis 3, we predicted that psychological detachment would moderate the positive within-person relationships of job boredom with job burnout, such that the relationship would weaken at higher levels of psychological detachment. Although we found that the moderation effect was statistically significant (see under Model 2 in Table 2), the effect was in the direction opposite to our prediction (see Fig. 2). The positive coefficient of the moderation term indicates that psychological detachment *reinforced* the relationship between job boredom and job burnout. Results of simple slope analysis demonstrated that simple slopes both at the mean level ($\gamma = 0.22$, $p = .045$) and at one standard deviation above the mean level of psychological detachment ($\gamma = 0.45$, $p = .001$) were statistically significant. Therefore, Hypothesis 3 was not supported.

In Hypothesis 4, we hypothesized that psychological detachment would moderate the positive within-person relationships of job boredom with turnover intentions, such that the relationship would weaken at higher levels of psychological detachment. As reported in Model 5 in Table 2, the moderation effect of psychological detachment on the job boredom–turnover intentions linkage was not statistically significant. Therefore, our findings did not provide support for Hypothesis 4.³

In Hypothesis 5, we predicted that perceived cyberloafing norms would moderate the positive within-person relationships of job boredom with job burnout, such that the relationship would weaken at a higher level of perceived cyberloafing norms. As shown in Model 3 in Table 2, we did not find a significant moderation effect of perceived cyberloafing norms on the job boredom–job burnout relationship, failing to support Hypothesis 5.

Lastly, in Hypothesis 6, we hypothesized that perceived cyberloafing norms would moderate the positive within-person relationships of job boredom with turnover intentions, such that the relationship would weaken at a higher level of perceived cyberloafing norms. As reported in Model 6 in Table 2, again, the moderation effect was not significant. Therefore, Hypothesis 6 was not supported.

² Conclusions from both Model 1 and Model 4 did not change when the lagged outcome variable was excluded from each model.

³ Due to a concern for multicollinearity between job boredom and psychological detachment, we computed estimates of variance inflation factor (VIF) for the relevant MLM models. The results showed that VIF was in the range of 1.02–1.08, suggesting that multicollinearity was not a concern.

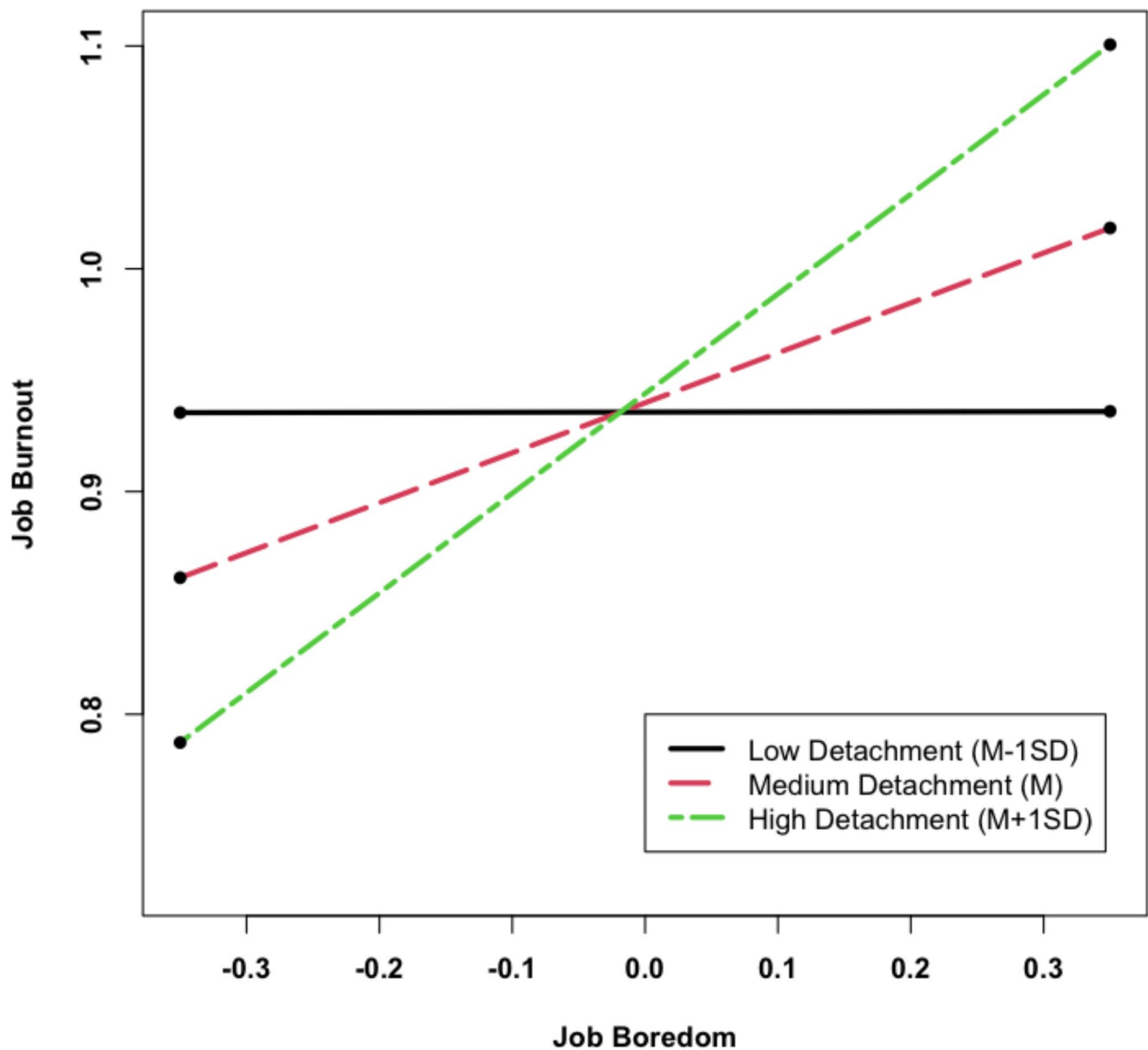


Fig. 2 The moderation of psychological detachment on the relationship between job boredom and job burnout. *Note.* The low, medium, and high cutoffs for psychological detachment were created at one

standard deviation below the mean, the mean, and one standard deviation above the mean, respectively

Discussion

As feeling bored can lead to an array of unpleasant consequences for employees and organizations, it is important for employees to learn how to cope with boredom at work and organizational leaders to help employees do so. Using a daily diary study over 10 consecutive workdays, we examined within-person fluctuations of job boredom and their relationship with job burnout and turnover intentions. Our study is among the first ESM studies in the job boredom literature to provide empirical evidence on the within-person associations of job boredom with job burnout and turnover

intentions. In addition, drawing from boredom reduction strategies proposed by the MAC model (Westgate & Wilson, 2018), we tested two moderators on the within-person relationships—psychological detachment from work during work hours and perceived cyberloafing norms at work.

The study findings show that feeling bored on a given day is associated with increased burnout on those days, which aligns with past findings at the between-person level. This suggests that job boredom fosters job burnout at both the within-person and between-person level (e.g., Harju et al., 2014; Sousa & Neves, 2021), supporting the E-R model (Meijman & Mulder, 1998) and confirming the

resource-draining effect of job boredom. Conversely, the finding that job boredom had a null association with turnover intentions at the within-person level contrasts with previous research reporting a positive association at the between-person level (e.g., Harju et al., 2014). One potential explanation is that the day-level impact of job boredom is not strong enough to change daily intentions to quit. That is, boredom makes the job more insufferable in terms of burnout, but it is not attributed as a reason to quit the job in that moment. Alternatively, it could be the case that the effect of job boredom was sufficiently large to significantly influence turnover intentions, but the transitory effect of boredom did not last until the timing of our turnover intentions assessment. This latter explanation is interesting and potentially warrants future research, because it begs the question of how long (or short) the effect of state boredom lasts. Because few guiding theories exist on this issue in the job boredom literature, we were unable to implement a theoretically-grounded timing of measurement for job boredom. However, our estimation of optimal time lags for the daily relationships of job boredom with job burnout and turnover intentions is 1.51 and 2.10 days, respectively (see Dormann & Griffin, 2015 for estimating optimal time lag). These estimates are very similar to the one-day time lag used in our study. Additionally, the daily measurement of job boredom in our study design followed a general temporal framework for job boredom measurement from previous ESM studies (e.g., Baratta & Spence, 2018; van Hooff & van Hooff, 2023).

Furthermore, our findings did not provide support for the restorative roles of psychological detachment, which is inconsistent with our prediction. Instead, we found that the deleterious impact of feeling bored at work on job burnout was *enhanced* on days that psychological detachment was high (in the early afternoon), compared to days that psychological detachment was low. This seems counter-intuitive, especially given recovery research showing that psychological detachment helps employees to recover from job demands (e.g., Sonnentag et al., 2010). One potential explanation concerns the fact that most studies on psychological detachment have studied detachment during *non-work* hours, whereas we measured detachment during *work* hours—namely, in the early afternoon. Given this difference, we conjecture that detachment during work (vs. non-work) hours may operate in a manner that is inconsistent with the conventional understanding of detachment during nonwork hours. This is manifested in part by the positive between-person correlation between psychological detachment and burnout in the current study (see Table 1), which is known to be negative in the case of detachment during non-work hours (Sonnentag et al., 2010). This harmful effect of detachment during work hours, however, does appear consistent with other research findings. For example, Zeschke

et al. (2025) found that detachment during work hours positively predicted emotional exhaustion both at the between- and within-person levels of analysis.

A second possibility is that detachment is conceptually somewhat similar to boredom, in that it could be an antecedent to, or an outcome of, boredom. It may be that detachment during work hours may not be an adaptive strategy but a maladaptive one, given that detachment showed positive within-person correlations with job boredom and turnover intentions. Another potential explanation is that the effect of detachment may have been confounded with *reattachment* (i.e., “a process during which employees mentally shift their attention to focus on work related matters and anticipate the upcoming workday or work shift,” Fritz & Taylor, 2020, p. 4). After a period of detachment at work, individuals must reattach to (or mentally reconnect with) work, and this reattachment process could be harmful to employee well-being.⁴ This reasoning suggests a boundary condition of detachment. Paradoxically, detaching from certain work activities—such as boring ones—could lead to lower rather than improved well-being, as individuals then must mentally re-engage with (i.e., reattach to) that same (boring) task. Furthermore, psychological detachment also did not moderate the relationship between job boredom and turnover intentions. One explanation may be that even if bored employees temporarily detach from work during work hours, turnover intentions could remain high for those who aspire to engage in meaningful work and pursue purposeful professional goals (Zacher & Frese, 2018).

Similarly, our results show that perceptions of social cues that internet-surfing is acceptable at work (i.e., perceived cyberloafing norms) may not help buffer the harmful impact on either job burnout or turnover intentions. This suggests that, in the case of job burnout, bored employees may experience burnout regardless of their perceptions of internet-use norms at work. Consistent with our findings, Panova and Lleras (2016) provided correlational evidence that suggest that information and communication technology use (e.g., using internet or smartphone) may not relate to mental health issues, which relate to burnout symptoms (Kelloway et al., 2023), among bored individuals. However, research on smartphone use does also suggest negative effects of cyberloafing on various well-being outcomes, such as anxiety, depression, stress, and sleep quality (e.g., Elhai et al., 2017; Stanković et al., 2021). In general, a further investigation is warranted as regards the two MAC model-derived strategies to reduce job boredom.

⁴ We would like to thank an anonymous reviewer for suggesting this insightful idea.

Practical implications

Our findings offer practical implications to HR practitioners. Specifically, our finding regarding the within-person effect of job boredom on burnout advises individual employees—and possibly HR managers to the extent relevant to current or future HR practices in their organizations—to be cognizant of the deleterious impact of daily boredom at work. Obviously, it would be almost impossible to completely avoid boredom at many jobs. Nonetheless, adopting day-to-day strategies for reducing boredom, such as job crafting (Harju et al., 2016a, b, 2018), peer support (Campagne, 2012), self-control (van Hooff & van Hooft, 2023), may help alleviate job burnout.

Regarding the null finding that perceived cyberloafing norms did not buffer the negative impact of job boredom, it may be premature to conclude that cyberloafing should be strictly restricted in the workplace. Although our finding did not provide support for the merit of cyberloafing, prior evidence shows that there are some benefits to allowing employees to cyberloaf to a certain extent, such as enhanced employee well-being (e.g., Lim & Chen, 2012; Wu et al., 2020a, b). When designing organizational policy or interventions for cyberloafing, however, organizational leaders should keep in mind that there certainly are risks of cyberloafing at work, as cyberloafing also has been considered a form of counterproductive work behavior (Lim & Teo, 2024). Therefore, we would advise HR managers and organizational leaders to take caution when designing cyberloafing-related policies or interventions in the workplace.

Limitations and future research directions

We also note some limitations with this study. First, because the measurement of the focal variables was only at the daily level, the interpretation of our findings is restricted to the daily assessment of job boredom and other focal variables. For pragmatic reasons of survey design (i.e., length of the survey), we were unable to measure all constructs in every ESM survey. Second, we only used self-report measures to assess the study variables, although self-ratings have been the most common approach to measure the focal constructs in prior studies. Nevertheless, future research is warranted to explore options for other rating sources to reduce common method variance (Podsakoff et al., 2003, 2024). Third, we measured perceived cyberloafing norms (vs. actual cyberloafing behavior). Future studies examining whether cyberloafing behavior would buffer the impact of job boredom would be helpful. Similarly, capturing detachment activities, instead of only the experience of detachment, would be valuable, given Westgate and Wilson's (2018) focus on behaviors per se. Also, we measured perceived

cyberloafing norms as an individual-level variable to test its between-person effect. Future research should consider assessing cyberloafing norms as a climate-/group-/team-level variable.

Future researchers should also consider additional moderators that help to alleviate the effect of job boredom. One potential moderator is occupational calling (for a review, see Dik & Duffy, 2009; Thompson & Bunderson, 2019), which can be defined as “a transcendent summons, experienced as originating beyond the self, to approach a particular life role (in this case work) in a manner oriented toward demonstrating or deriving a sense of purpose or meaningfulness and that holds other-oriented values and goals as primary sources of motivation” (Dik & Duffy, 2009, p.427). Research on calling suggests that individuals who see work as a calling tend to have higher job satisfaction, experience less feelings of burnout, and harbor less thoughts of quitting (Thompson & Bunderson, 2019). Because workers who perceive their jobs as a calling tend to respond adaptively to work-related challenges (Schabram & Maitlis, 2017), research generally suggests that calling alleviates the effects of negative organizational phenomena, such as burnout (Creed et al., 2014), emotional exhaustion (Rawat & Nadavulakere, 2015), and health and health satisfaction (Wrzesniewski et al., 1997). Given the advantages of calling, we speculate that perceiving/having a calling may help reduce the effects of job boredom.

Furthermore, researchers may also consider the effects of job design and job crafting on job boredom. In one relevant three-year cross-lagged panel study, Harju et al. (2016a, b) suggest that increasing challenges at work (i.e., a dimension of job crafting) can reduce subsequent boredom; however, they also note that job boredom hampers job crafting. Additionally, using a control-process model of coping, van Hooff and van Hooft (2023) show that job crafting in the morning may be unrelated to job boredom in the afternoon. These findings suggest that more in-depth examination of job boredom as an outcome of job crafting may be useful. Conversely, future research also may benefit from examining job boredom as a predictor of job crafting, given that boredom can motivate individuals to modify the current task/situation to be more enjoyable or interesting or to engage in another task that is more enjoyable or interesting (Westgate & Wilson, 2018). Overall, the bidirectional relationship between job boredom and job crafting seems to warrant further investigation. Relatedly, future researchers also should consider helping behavior (see Bolino & Grant, 2016, for review) as an alternative strategy to reduce boredom at work. Given that bored individuals may help others to alleviate boredom (van Tilburg & Igou, 2017), helping behavior can be another boredom reduction strategy worth future examination.

Lastly, even though job burnout is a multi-dimensional construct (Maslach et al., 2001), we used a single-item measure to assess overall job burnout due to the practical constraints with survey length. With a more nuanced approach, future researchers should study the within-person effect of job boredom on the three dimensions of job burnout: exhaustion, cynicism, and sense of inefficacy (Maslach et al., 2001).

Conclusion

We extend the literature on job boredom by examining within-person effects of job boredom on workplace outcomes (i.e., burnout and turnover intentions) that relate to important practical implications. Using an ESM study over 10 consecutive workdays, we found that job boredom was positively related to burnout at the within-person level, whereas job boredom was not related to turnover intentions. We did not find support for the hypothesized restorative role of either psychological detachment during work hours or perceived cyberloafing norms. We hope that the findings of this study provide researchers and practitioners with an empirical ground for understanding job boredom at the within-person level.

Acknowledgements We thank Sydney E. Farrar for assisting with survey creation and data collection.

Funding This work was supported in part by a research award to the first author from George Mason University.

Data availability The data and analysis syntax used in this study are available in the following online data repository: <https://doi.org/10.17605/OSF.IO/UDR9A>.

Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

Ethics approval This research was approved (as exempt) by the George Mason University Institutional Review Board (IRB approval #1890536-1).

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright

holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Anandarajan, M., Simmers, C., & Igbaria, M. (2000). An exploratory investigation of the antecedents and impact of internet usage: An individual perspective. *Behaviour & Information Technology*, 19(1), 69–85. <https://doi.org/10.1080/014492900118803>
- Antoun, C., Zhang, C., Conrad, F. G., & Schober, M. F. (2016). Comparisons of online recruitment strategies for convenience samples: Craigslist, Google AdWords, Facebook, and Amazon mechanical Turk. *Field Methods*, 28(3), 231–246. <https://doi.org/10.1177/1525822X15603149>
- Bakker, A. B., ten Brummelhuis, L. L., Prins, J. T., & van der Heijden, F. M. M. A. (2011). Applying the job demands-resources model to the work-home interface: A study among medical residents and their partners. *Journal of Vocational Behavior*, 79(1), 170–180. <https://doi.org/10.1016/j.jvb.2010.12.004>
- Bakker, A. B., Demerouti, E., & Sanz-Vergel, A. (2023). Job demands–resources theory: Ten years later. *Annual Review of Organizational Psychology and Organizational Behavior*, 10(1), 25–53. <https://doi.org/10.1146/annurev-orgpsych-120920-053933>
- Baratta, P. L., & Spence, J. R. (2018). Capturing the noonday demon: Development and validation of the state boredom inventory. *European Journal of Work and Organizational Psychology*, 27(4), 477–492. <https://doi.org/10.1080/1359432X.2018.1481830>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bench, S. W., & Lench, H. C. (2013). On the function of boredom. *Behavioral Sciences*, 3(3), 459–472. <https://doi.org/10.3390/bs3030459>
- Bench, S. W., & Lench, H. C. (2019). Boredom as a seeking state: Boredom prompts the pursuit of novel (even negative) experiences. *Emotion*, 19(2), 242–254. <https://doi.org/10.1037/emo0000433>
- Blanco-Donoso, L. M., Moreno-Jiménez, J., Hernández-Hurtado, M., Cifri-Gavela, J. L., Jacobs, S., & Garrosa, E. (2021). Daily work-family conflict and burnout to explain the leaving intentions and vitality levels of healthcare workers: Interactive effects using an experience-sampling method. *International Journal of Environmental Research and Public Health*, 18(4), 1932. <https://doi.org/10.3390/ijerph18041932>
- Bolino, M. C., & Grant, A. M. (2016). The bright side of being prosocial at work, and the dark side, too: A review and agenda for research on other-oriented motives, behavior, and impact in organizations. *Academy of Management Annals*, 10(1), 599–670. <https://doi.org/10.5465/19416520.2016.1153260>
- Brodsky, A., & Amabile, T. M. (2018). The downside of downtime: The prevalence and workpacing consequences of idle time at work. *Journal of Applied Psychology*, 103(5), 496–512. <https://doi.org/10.1037/apl0000294>
- Bruursema, K., Kessler, S. R., & Spector, P. E. (2011). Bored employees misbehaving: The relationship between boredom and counterproductive work behaviour. *Work & Stress*, 25(2), 93–107. <https://doi.org/10.1080/02678373.2011.596670>
- Campagne, D. M. (2012). When therapists run out of steam: Professional boredom or burnout? *Revista de Psicopatología y Psicología Clínica*, 17(1), 75–85.
- Chin, A., Markey, A., Bhargava, S., Kassam, K. S., & Loewenstein, G. (2017). Bored in the USA: Experience sampling and boredom in

- everyday life. *Emotion*, 17(2), 359–368. <https://doi.org/10.1037/emo0000232>
- Cordes, C. L., & Dougherty, T. W. (1993). A review and an integration of research on job burnout. *Academy of Management Review*, 18(4), 621–656. <https://doi.org/10.5465/amr.1993.9402210153>
- Creed, P. A., Rogers, M. E., Praskova, A., & Searle, J. (2014). Career calling as a personal resource moderator between environmental demands and burnout in Australian junior Doctors. *Journal of Career Development*, 41(6), 547–561. <https://doi.org/10.1177/0894845313520493>
- Cummings, M. L., Gao, F., & Thornburg, K. M. (2016). Boredom in the workplace: A new look at an old problem. *Human Factors*, 58(2), 279–300. <https://doi.org/10.1177/0018720815609503>
- Dik, B. J., & Duffy, R. D. (2009). Calling and vocation at work: Definitions and prospects for research and practice. *The Counseling Psychologist*, 37(3), 424–450. <https://doi.org/10.1177/0011000008316430>
- Dormann, C., & Griffin, M. A. (2015). Optimal time lags in panel studies. *Psychological Methods*, 20(4), 489–505. <https://doi.org/10.1037/met0000041>
- Dunford, B. B., Shipp, A. J., Boss, R. W., Angermeier, I., & Boss, A. D. (2012). Is burnout static or dynamic? A career transition perspective of employee burnout trajectories. *Journal of Applied Psychology*, 97(3), 637–650. <https://doi.org/10.1037/a0027060>
- Eastwood, J. D., Frischen, A., Fenske, M. J., & Smilek, D. (2012). The unengaged Mind: Defining boredom in terms of attention. *Perspectives on Psychological Science*, 7(5), 482–495. <https://doi.org/10.1177/1745691612456044>
- Elhai, J. D., Dvorak, R. D., Levine, J. C., & Hall, B. J. (2017). Problematic smartphone use: A conceptual overview and systematic review of relations with anxiety and depression psychopathology. *Journal of Affective Disorders*, 207, 251–259. <https://doi.org/10.1016/j.jad.2016.08.030>
- Elpidorou, A. (2017). The good of boredom. *Philosophical Psychology*, 31(3), 323–351. <https://doi.org/10.1080/09515089.2017.1346240>
- Elpidorou, A. (2018). The bored mind is a guiding mind: Toward a regulatory theory of boredom. *Phenomenology and the Cognitive Sciences*, 17(3), 455–484. <https://doi.org/10.1007/s11097-017-9515-1>
- Farmer, R., & Sundberg, N. D. (1986). Boredom proneness—the development and correlates of a new scale. *Journal of Personality Assessment*, 50(1), 4–17. https://doi.org/10.1207/s15327752jpa5001_2
- Fisher, C. D. (1993). Boredom at work: A neglected concept. *Human Relations*, 46(3), 395–417. <https://doi.org/10.1177/001872679304600305>
- Fisher, G. G., Matthews, R. A., & Gibbons, A. M. (2016). Developing and investigating the use of single-item measures in organizational research. *Journal of Occupational Health Psychology*, 21(1), 3–23. <https://doi.org/10.1037/a0039139>
- Freudenberger, H. J. (1974). Staff burn-out. *Journal of Social Issues*, 30(1), 159–165. <https://doi.org/10.1111/j.1540-4560.1974.tb00706.x>
- Fritz, C., & Taylor, M. (2020). Detaching from and reattaching to work: The dance of work-nonwork balance. *Organizational Dynamics*, 49(4). <https://doi.org/10.1016/j.orgdyn.2019.04.010>
- Gabriel, A. S., Podsakoff, N. P., Beal, D. J., Scott, B. A., Sonnentag, S., Trougakos, J. P., & Butts, M. M. (2019). Experience sampling methods: A discussion of critical trends and considerations for scholarly advancement. *Organizational Research Methods*, 22(4), 969–1006. <https://doi.org/10.1177/1094428118802626>
- Gaines, J., & Jermier, J. M. (1983). Emotional exhaustion in a high stress organization. *Academy of Management Journal*, 26(4), 567–586. <https://doi.org/10.5465/255907>
- Game, A. M. (2007). Workplace boredom coping: Health, safety, and HR implications. *Personnel Review*, 36(5), 701–721. <https://doi.org/10.1108/00483480710774007>
- Goetz, T., Frenzel, A. C., Hall, N. C., Nett, U. E., Pekrun, R., & Lipnevich, A. A. (2014). Types of boredom: An experience sampling approach. *Motivation and Emotion*, 38(3), 401–419. <https://doi.org/10.1007/s11031-013-9385-y>
- González-Romá, V., & Hernández, A. (2017). Multilevel modeling: Research-based lessons for substantive researchers. *Annual Review of Organizational Psychology and Organizational Behavior*, 4, 183–210. <https://doi.org/10.1146/annurev-orgpsych-041015-062407>
- Harju, L., Hakanen, J. J., & Schaufeli, W. B. (2014). Job boredom and its correlates in 87 Finnish organizations. *Journal of Occupational and Environmental Medicine*, 56(9), 911–918. <https://doi.org/10.1097/JOM.0000000000000248>
- Harju, L. K., Hakanen, J. J., & Schaufeli, W. B. (2016a). Can job crafting reduce job boredom and increase work engagement? A three-year cross-lagged panel study. *Journal of Vocational Behavior*, 95–96, 11–20. <https://doi.org/10.1016/j.jvb.2016.07.001>
- Harju, L. K., Hakanen, J. J., & Schaufeli, W. B. (2016b). Can job crafting reduce job boredom and increase work engagement? A three-year cross-lagged panel study. *Journal of Vocational Behavior*, 95, 11–20. <https://doi.org/10.1016/j.jvb.2016.07.001>
- Harju, L. K., Schaufeli, W. B., & Hakanen, J. J. (2018). A multilevel study on servant leadership, job boredom and job crafting. *Journal of Managerial Psychology*, 33(1), 2–14. <https://doi.org/10.1108/JMP-08-2016-0237>
- Harju, L. K., Van Hootegeem, A., & De Witte, H. (2022). Bored or burning out? Reciprocal effects between job stressors, boredom and burnout. *Journal of Vocational Behavior*, 139, 103807. <https://doi.org/10.1016/j.jvb.2022.103807>
- Hom, P. W., Lee, T. W., Shaw, J. D., & Hausknecht, J. P. (2017). One hundred years of employee turnover theory and research. *Journal of Applied Psychology*, 102(3), 530–545. <https://doi.org/10.1037/apl0000103>
- Julian, M. W. (2001). The consequences of ignoring multilevel data structures in nonhierarchical covariance modeling. *Structural Equation Modeling*, 8(3), 325–352. https://doi.org/10.1207/S15328007SEM0803_1
- Kass, S. J., Vodanovich, S. J., & Callender, A. (2001). State-trait boredom: Relationship to absenteeism, tenure, and job satisfaction. *Journal of Business and Psychology*, 16, 317–327. <https://doi.org/10.1023/A:1011121503118>
- Kelloway, E. K., Dimoff, J. K., & Gilbert, S. (2023). Mental health in the workplace. *Annual Review of Organizational Psychology and Organizational Behavior*, 10, 363–387. <https://doi.org/10.1146/annurev-orgpsych-120920-050527>
- Kim, J., Park, J., Sohn, Y. W., & Lim, J. I. (2021). Perceived overqualification, boredom, and extra-role behaviors: Testing a moderated mediation model. *Journal of Career Development*, 48(4), 400–414. <https://doi.org/10.1177/0894845319853879>
- Kim, J. J., Kaplan, S. A., Aitken, J. A., & Ponce, L. P. (2024). Within-person dynamics of job boredom and counterproductive work behavior: A latent change score modeling approach. *Affective Science*, 5(3), 273–279. <https://doi.org/10.1007/s42761-024-00256-y>
- Kim, S., Park, Y., & Headrick, L. (2015). Employees' micro-break activities and job performance: An examination of telemarketing employees. In *Academy of management proceedings* (Vol. 2015, No. 1, p. 13943). Academy of Management. <https://doi.org/10.5465/ambpp.2015.169>
- Kim, S., Park, Y., & Niu, Q. (2017). Micro-break activities at work to recover from daily work demands. *Journal of Organizational Behavior*, 38(1), 28–44. <https://doi.org/10.1002/job.2109>

- Koay, K. Y., & Soh, P. C. H. (2018). Should cyberloafing be allowed in the workplace? *Human Resource Management International Digest*, 26(7), 4–6. <https://doi.org/10.1108/HRMID-05-2018-0107>
- Lee, T. W., Burch, T. C., & Mitchell, T. R. (2014). The story of why we stay: A review of job embeddedness. *Annual Review of Organizational Psychology and Organizational Behavior*, 1(1), 199–216. <https://doi.org/10.1146/annurev-orgpsych-031413-091244>
- Liberman, B., Seidman, G., McKenna, K. Y., & Buffardi, L. E. (2011). Employee job attitudes and organizational characteristics as predictors of cyberloafing. *Computers in Human Behavior*, 27(6), 2192–2199. <https://doi.org/10.1016/j.chb.2011.06.015>
- Lim, V. K. (2002). The IT way of loafing on the job: Cyberloafing, neutralizing and organizational justice. *Journal of Organizational Behavior*, 23(5), 675–694. <https://doi.org/10.1002/job.161>
- Lim, V. K., & Chen, D. J. (2012). Cyberloafing at the workplace: Gain or drain on work? *Behaviour & Information Technology*, 31(4), 343–353. <https://doi.org/10.1080/01449290903353054>
- Lim, V. K., & Teo, T. S. (2024). Cyberloafing: A review and research agenda. *Applied Psychology*, 73(1), 441–484. <https://doi.org/10.1111/apps.12452>
- Loukidou, L., Loan-Clarke, J., & Daniels, K. (2009). Boredom in the workplace: More than monotonous tasks. *International Journal of Management Reviews*, 11(4), 381–405. <https://doi.org/10.1111/j.1468-2370.2009.00267.x>
- Mael, F., & Jex, S. (2015). Workplace boredom: An integrative model of traditional and contemporary approaches. *Group & Organization Management*, 40(2), 131–159. <https://doi.org/10.1177/1059601115575148>
- Maslach, C., Schaufeli, W. B., & Leiter, M. P. (2001). Job burnout. *Annual Review of Psychology*, 52(1), 397–422. <https://doi.org/10.1146/annurev.psych.52.1.397>
- Mason, W., & Suri, S. (2012). Conducting behavioral research on Amazon's mechanical Turk. *Behavior Research Methods*, 44(1), 1–23. <https://doi.org/10.3758/s13428-011-0124-6>
- McCormick, B. W., Reeves, C. J., Downes, P. E., Li, N., & Ilies, R. (2020). Scientific contributions of within-person research in management: Making the juice worth the squeeze. *Journal of Management*, 46(2), 321–350. <https://doi.org/10.1177/0149206318788435>
- Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods*, 17(3), 437–455. <https://doi.org/10.1037/a0028085>
- Meijman, T. F., & Mulder, G. (1998). Psychological aspects of workload. In C. J. De Wolff, P. J. D. Drenth, & T. Henk (Eds.), *Handbook of work and organizational psychology* (Volume 2, pp. 5–33). Psychology Press. <https://doi.org/10.4324/9780203765425>
- Melamed, S., Ben-Avi, I., Luz, J., & Green, M. S. (1995). Objective and subjective work monotony: Effects on job satisfaction, psychological distress, and absenteeism in blue-collar workers. *Journal of Applied Psychology*, 80(1), 29–42. <https://doi.org/10.1037/0021-9010.80.1.29>
- Mikulas, W. L., & Vodanovich, S. J. (1993). The essence of boredom. *The Psychological Record*, 43(1), 3–12.
- Miller, D. T., & Prentice, D. A. (2016). Changing norms to change behavior. *Annual Review of Psychology*, 67(1), 339–361. <https://doi.org/10.1146/annurev-psych-010814-015013>
- O'Hanlon, J. F. (1981). Boredom: Practical consequences and a theory. *Acta Psychologica*, 49(1), 53–82. [https://doi.org/10.1016/0001-6918\(81\)90033-0](https://doi.org/10.1016/0001-6918(81)90033-0)
- Panova, T., & Lleras, A. (2016). Avoidance or boredom: Negative mental health outcomes associated with use of information and communication technologies depend on users' motivations. *Computers in Human Behavior*, 58, 249–258. <https://doi.org/10.1016/j.chb.2015.12.062>
- Park, G., Lim, B. C., & Oh, H. S. (2019). Why being bored might not be a bad thing after all. *Academy of Management Discoveries*, 5(1), 78–92. <https://doi.org/10.5465/amd.2017.0033>
- Pindek, S., Krajcevska, A., & Spector, P. E. (2018). Cyberloafing as a coping mechanism: Dealing with workplace boredom. *Computers in Human Behavior*, 86, 147–152. <https://doi.org/10.1016/j.chb.2018.04.040>
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879–903. <https://doi.org/10.1037/0021-9010.88.5.879>
- Podsakoff, P. M., Podsakoff, N. P., Williams, L. J., Huang, C., & Yang, J. (2024). Common method bias: It's bad, it's complex, it's widespread, and it's not easy to fix. *Annual Review of Organizational Psychology and Organizational Behavior*, 11(1), 17–61. <https://doi.org/10.1146/annurev-orgpsych-110721-040030>
- Rawat, A., & Nadavulakere, S. (2015). Examining the outcomes of having a calling: Does context matter? *Journal of Business and Psychology*, 30, 499–512. <https://doi.org/10.1007/s10869-014-9378-1>
- Reijseger, G., Schaufeli, W. B., Peeters, M. C., Taris, T. W., Van Beek, I., & Ouwenel, E. (2013). Watching the paint dry at work: Psychometric examination of the Dutch boredom scale. *Anxiety Stress & Coping*, 26(5), 508–525. <https://doi.org/10.1080/10615806.2012.720676>
- Rimal, R. N., & Real, K. (2003). Understanding the influence of perceived norms on behaviors. *Communication Theory*, 13(2), 184–203. <https://doi.org/10.1111/j.1468-2885.2003.tb00288.x>
- Rimal, R. N., & Real, K. (2005). How behaviors are influenced by perceived norms: A test of the theory of normative social behavior. *Communication Research*, 32(3), 389–414. <https://doi.org/10.1177/0093650205275385>
- Schabram, K., & Maitlis, S. (2017). Negotiating the challenges of a calling: Emotion and enacted sensemaking in animal shelter work. *Academy of Management Journal*, 60(2), 584–609. <https://doi.org/10.5465/amj.2013.0665>
- Shi, X., Gordon, S., & Tang, C. H. (2021). Momentary well-being matters: Daily fluctuations in hotel employees' turnover intention. *Tourism Management*, 83, 104212. <https://doi.org/10.1016/j.tourman.2020.104212>
- Shockley, K. M., Allen, T. D., Dodd, H., & Waiwood, A. M. (2021). Remote worker communication during COVID-19: The role of quantity, quality, and supervisor expectation-setting. *Journal of Applied Psychology*, 106(10), 1466–1482. <https://doi.org/10.1037/apl0000970>
- Sischka, P. E., Décieux, J. P., Mergener, A., Neufang, K. M., & Schmidt, A. F. (2022). The impact of forced answering and reactance on answering behavior in online surveys. *Social Science Computer Review*, 40(2), 405–425. <https://doi.org/10.1177/0894439320907067>
- Smith, C. A., & Ellsworth, P. C. (1985). Patterns of cognitive appraisal in emotion. *Journal of Personality and Social Psychology*, 48(4), 813–838. <https://doi.org/10.1037/0022-3514.48.4.813>
- Sonnentag, S., Binnewies, C., & Mojza, E. J. (2010). Staying well and engaged when demands are high: The role of psychological detachment. *Journal of Applied Psychology*, 95(5), 965–976. <https://doi.org/10.1037/a0020032>
- Sonnentag, S., & Fritz, C. (2007). The recovery experience questionnaire: Development and validation of a measure for assessing recuperation and unwinding from work. *Journal of Occupational Health Psychology*, 12(3), 204–221. <https://doi.org/10.1037/1076-8998.12.3.204>
- Sousa, T., & Neves, P. (2021). Two Tales of rumination and burnout: Examining the effects of boredom and overload. *Applied Psychology*, 70(3), 1018–1044. <https://doi.org/10.1111/apps.12257>

- Spanouli, A., Hofmans, J., & Dalal, R. S. (2023). Coping with daily boredom: Exploring the relationships of job boredom, counter-productive work behavior, organizational citizenship behavior, and cognitive reappraisal. *Motivation and Emotion*, 47(5), 810–827. <https://doi.org/10.1007/s11031-023-10017-2>
- Stanković, M., Nešić, M., Čičević, S., & Shi, Z. (2021). Association of smartphone use with depression, anxiety, stress, sleep quality, and internet addiction. Empirical evidence from a smartphone application. *Personality and Individual Differences*, 168, 110342. <https://doi.org/10.1016/j.paid.2020.110342>
- Stoffregen, T. A. (2003). Affordances as properties of the animal–environment system. *Ecological Psychology*, 15(2), 115–134. https://doi.org/10.1207/S15326969ECO1502_2
- Thompson, J. A., & Bunderson, J. S. (2019). Research on work as a calling... and how to make it matter. *Annual Review of Organizational Psychology and Organizational Behavior*, 6, 421–443. <https://doi.org/10.1146/annurev-orgpsych-012218-015140>
- Turk, T., Elhady, M. T., Rashed, S., Abdelkhalek, M., Nasef, S. A., Khallaf, A. M., Mohammed, A. T., Attia, A. W., Adhikari, P., Amin, M. A., Hirayama, K., & Huy, N. T. (2018). Quality of reporting web-based and non-web-based survey studies: What authors, reviewers and consumers should consider. *PloS One*, 13(6), e0194239. <https://doi.org/10.1371/journal.pone.0194239>
- van Hooff, M. L. M., & van Hooft, E. A. J. (2014). Boredom at work: Proximal and distal consequences of affective work-related boredom. *Journal of Occupational Health Psychology*, 19(3), 348–359. <https://doi.org/10.1037/a0036821>
- van Hooff, M. L., & van Hooft, E. A. (2016). Work-related boredom and depressed mood from a daily perspective: The moderating roles of work centrality and need satisfaction. *Work & Stress*, 30(3), 209–227. <https://doi.org/10.1080/02678373.2016.1206151>
- van Hooff, M. L., & van Hooft, E. A. (2017). Boredom at work: Towards a dynamic spillover model of need satisfaction, work motivation, and work-related boredom. *European Journal of Work and Organizational Psychology*, 26(1), 133–148. <https://doi.org/10.1080/1359432X.2016.1241769>
- van Hooff, M. L., & van Hooft, E. A. (2023). Dealing with daily boredom at work: Does self-control explain who engages in distractive behaviour or job crafting as a coping mechanism? *Work & Stress*, 37(2), 248–268. <https://doi.org/10.1080/02678373.2022.2129515>
- van Tilburg, W. A., & Igou, E. R. (2017). Can boredom help? Increased prosocial intentions in response to boredom. *Self and Identity*, 16(1), 82–96. <https://doi.org/10.1080/15298868.2016.1218925>
- Velasco, F. (2017). Understanding workplace boredom among service employees: Qualitative insights and employee outcomes. *Journal of Managerial Issues*, 29(3), 278–293. <https://www.jstor.org/stable/45176551>
- Virgă, D., Pattusamy, M., & Kumar, D. P. (2022). How psychological capital is related to academic performance, burnout, and boredom? The mediating role of study engagement. *Current Psychology*, 41, 6731–6743. <https://doi.org/10.1007/s12144-020-01162-9>
- Wan, H. C., Downey, L. A., & Stough, C. (2014). Understanding non-work presenteeism: Relationships between emotional intelligence, boredom, procrastination and job stress. *Personality and Individual Differences*, 65, 86–90. <https://doi.org/10.1016/j.paid.2014.01.018>
- Watson, D., & Tellegen, A. (1985). Toward a consensual structure of mood. *Psychological Bulletin*, 98(2), 219–235. <https://doi.org/10.1037/0033-2909.98.2.219>
- Weinstein, L., Xie, X., & Cleanthous, C. C. (1995). Purpose in life, boredom, and volunteerism in a group of retirees. *Psychological Reports*, 76(2), 482–482. <https://doi.org/10.2466/pr0.1995.76.2.482>
- Westgate, E. C., & Wilson, T. D. (2018). Boring thoughts and bored minds: The MAC model of boredom and cognitive engagement. *Psychological Review*, 125(5), 689–713. <https://doi.org/10.1037/rev0000097>
- Wheeler, L., & Reis, H. T. (1991). Self-recording of everyday life events: Origins, types, and uses. *Journal of Personality*, 59(3), 339–354. <https://doi.org/10.1111/j.1467-6494.1991.tb00252.x>
- Wrzesniewski, A., McCauley, C., Rozin, P., & Schwartz, B. (1997). Jobs, careers, and callings: People's relations to their work. *Journal of Research in Personality*, 31(1), 21–33. <https://doi.org/10.1006/jrpe.1997.2162>
- Wu, J., Mei, W., Liu, L., & Ugrin, J. C. (2020a). The bright and dark sides of social cyberloafing: Effects on employee mental health in China. *Journal of Business Research*, 112, 56–64. <https://doi.org/10.1016/j.jbusres.2020.02.043>
- Wu, J., Mei, W., Ugrin, J., Liu, L., & Wang, F. (2020b). Curvilinear performance effects of social cyberloafing out of class: The mediating role as a recovery experience. *Information Technology & People*, 34(2), 581–598. <https://doi.org/10.1108/ITP-03-2019-0105>
- Zacher, H., & Frese, M. (2018). Action regulation theory: Foundations, current knowledge, and future directions. In N. Anderson, D. S. Ones, C. Viswesvaran, & H. K. Sinangil (Eds.), *The SAGE handbook of industrial, work, & organizational psychology* (2nd ed., Vol. 2, pp. 122–144). <https://doi.org/10.4135/9781473914957.n7>
- Zeschke, M., & Zacher, H. (2023). Is it bad because it is boring? Effects of idle time on employee outcomes. *European Journal of Work and Organizational Psychology*, 32(6), 886–901. <https://doi.org/10.1080/1359432X.2023.2261661>
- Zeschke, M., Venz, L., & Zacher, H. (2025). Relations between idle time, exhaustion, and engagement at work: The role of work overload, autonomy, and recovery experiences. *International Journal of Stress Management*. Advance online publication. <https://doi.org/10.1037/str0000340>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.